Expanded Standard MPC vs Project FIR-QP-MPC

A. Standard Receding-Horizon MPC

State measurement / estimation
$\hat{x}_k$

↓

Reference and constraint update
$r_{k:k+N}, \quad x \in X, \quad u \in U$

↓

Horizon prediction
$X = F(\hat{x}_k, U)$

↓

Objective construction
tracking + control effort + terminal cost
$J = \Sigma \|x\text{-}r\|_Q^2 + \|u\|_R^2 + V_f$

↓

QP / NLP solution
$U_k^* = arg\ min_U J(\hat{x}_k, U)$

↓

Apply the first optimal input
$u_k = u_{0|k}^*$

↓

Plant evolution; advance one sample
$x_{k+1} = f(x_k, u_k)$

B. Project FIR-QP-MPC

Offline nominal references
FDDP nominal + C0-aligned pitch reference
$q_{ref}(t), \quad \theta_{ref}(t)$

↓

Online state and contact measurement
$\theta_k, \quad \theta_k, \quad toe/heel\ contact$

↓

Contact-event phase alignment
$t_\varphi = clip(t + \Delta t_\varphi, 0, T)$

↓

Reference-aligned error and known-response separation
$e_k = \theta_k - \theta_{ref}(t_\varphi)$
$d_k = e_k - \Sigma h_i \alpha_{k-i}$

↓

AR natural response + FIR command-response forecast
$E = E_{free} + SA$

↓

Amplitude / slew constrained QP; first correction
$A^* = arg\ min\ q\|E_{free}+SA\|^2 + r\|A\|^2 + s\|DA\text{-}d_{prev}\|^2$
$\alpha_k = A_0^*, \quad \Delta q_k = G\alpha_k$

↓

Position command and actuator proxy
$q_{cmd} = q_{ref}(t_\varphi) + \Delta q_k$

Both pipelines re-predict, re-optimise, and apply only the first action at each sample.
τ_{ref}
is analysis-only; the runtime output is the position command
q_{cmd}
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